

Row Space, Column Space, Null Space, Left Null Space :-

Row Space, Column Space, Null Space & Left Null Space

The Four Fundamental Subspaces

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graph TD; A[The Four Fundamental Subspaces] --> B[Column Space]; A --> C[Row Space]; A --> D[Nullspace]; A --> E[Left Nullspace];
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→ vector space is set (collection) of vectors such that their Linear combination will give a vector from the same set
Ex: $\mathbb{R}^2, \mathbb{R}^3, \mathbb{R}^4 \dots \mathbb{R}^n$

Subspace = Subset + Vector Space

→ For a set of vectors, Span is the set of all possible linear combination of vectors.

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→ Basis of a vector space is a minimal set of vectors that

- ① fills (spans) the space
- ② is linearly Independent

* properties not impacted by elementary Row transformation of a Matrix

① Rank

$$\text{Rank}(A) = \text{Rank}(\text{REF}(A))$$

NOTE

$\text{REF}(A)$ = Row echelon form of A

② Linear relationship among rows

dependence Independence

③ Row space

span of rows remains the same

④ Nullity

Determinant value, EigenValue, Column space etc changes on elementary row operations

⑤ Solution set of $Ax=b$

⑥ Pivot Columns

⑦ Singular vs Nonsingular nature

⑧ dimension of column space

→ Therefore at many places instead of working with A, we echelon form of A or row reduced echelon form of A.

For any Matrix $A_{m \times n}$, the number of Linearly Independent rows and number of Linearly Independent columns are equal

→ In Gaussian elimination, we do elementary row transformation and reduce the matrix to row-echelon form.

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow[\substack{R_2 = R_2 - 2R_1 \\ R_3 = R_3 - R_1}]{\substack{R_2 = R_2 - 2R_1 \\ R_3 = R_3 - R_1}} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -2 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

→ Also Linear dependency among columns remain preserved during elementary row operations.

→ In row echelon form no of non-zero row is equal to no of linearly independent rows, and each non-zero row contains a pivot.

↳ Therefore $\# \text{ Linearly Independent rows} = \# \text{ Linearly Independent Columns. } (A)$

* Intuitive meaning of $Ax = b$

$$\begin{aligned} 2x + 4y - 2z &= 2 \\ 4x + 9y - 3z &= 8 \\ -2x - 3y + 7z &= 10 \end{aligned} \iff \begin{bmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 8 \\ 10 \end{bmatrix}$$

System of Linear equations

Matrix form: $Ax = b$

→ find a column vector x such that the Linear combination of columns of A is equal to column vector b .

↳ Column Space

→ Let A be a matrix consisting of n column vectors

$$A = \begin{bmatrix} a_1 & a_2 & a_3 & \dots & a_n \end{bmatrix}_{m \times n}$$

→ column space for matrix A is denoted by $\text{Col}(A)$ or $C(A)$

Column Space of a matrix is the set of all vectors that can be expressed as a linear combination of its columns

→ $\text{Col}(A) = x_1 a_1 + x_2 a_2 + x_3 a_3 + \dots + x_n a_n$, $x_1, x_2, \dots, x_n$ are scalars

$$= \begin{bmatrix} a_1 & a_2 & a_3 & \dots & a_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \{Ax : x \in \mathbb{R}^n\}$$

$$\text{Col}(A) = \text{Span}(\{a_1, a_2, a_3, \dots, a_n\})$$

↳ Since $A \in \mathbb{R}^{m \times n}$

① Each column is in $\mathbb{R}^m$ and total n columns.

② So in best case we can have m or more Linearly independent columns → It will span entire $\mathbb{R}^m$

③ And in worst case we will have 1 Linearly independent columns.
↳ It will span a line $\mathbb{R}^1$

Therefore column space is a subspace of $\mathbb{R}^m$

→ Dimension of column space = rank = no of pivot columns = r

Example: Analyze column space of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 1 & 2 & 5 \end{bmatrix}$

$$\rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 1 & 2 & 5 \end{bmatrix} \xrightarrow[R_3 = R_3 - R_1]{R_2 = R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 2 \end{bmatrix} \xrightarrow{R_3 = R_3 - 2R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \text{ Row Echelon Form}$$

↳ no of pivots in row echelon form = 2

Therefore $\text{rank}(A) = 2 \leftarrow$ only 2 out of 3 columns of A are linearly independent

↳ $\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \\ 5 \end{bmatrix} \right\}$ are linearly independent columns and $\begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}$ is LD on c_1 and c_3 .

$$\rightarrow \text{Col}(A) = \text{span} \left(\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \\ 5 \end{bmatrix} \right\} \right) = \text{span} \left(\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \\ 5 \end{bmatrix} \right\} \right)$$

↳ basis for column space = $\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \\ 5 \end{bmatrix} \right\}$ ← Take columns from A corresponding to pivot columns.

↳ dimension of column space = rank = 2

Row space

→ Let A be a matrix consisting of m row vectors $A = \begin{bmatrix} -r_1- \\ -r_2- \\ \vdots \\ -r_m- \end{bmatrix}$

→ Row space for matrix A is denoted by $\text{Row}(A)$ or $\text{Col}(A^T)$ ↑
Column space of A^T

Row Space of a matrix is the set of all vectors that can be expressed as a linear combination of its rows

→ $\text{Col}(A) = x_1 r_1 + x_2 r_2 + x_3 r_3 + \dots + x_n r_n$, $x_1, x_2, \dots, x_n$ are scalars

$$= [r_1 \ r_2 \ r_3 \ \dots \ r_n] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \{A^T x : x \in \mathbb{R}^n\}$$

$$\text{Row}(A) = \text{Span}(\{r_1, r_2, r_3, \dots, r_n\})$$

→ $\text{Row}(A) = \text{Col}(A^T) \subseteq \mathbb{R}^n$

Row space is a subspace of $\mathbb{R}^n$

→ dimension of Row Space is equal to rank of A.

$$\dim(\text{Row}(A)) = \text{Rank}(A) = \text{No of pivot columns} = r$$

Null Space

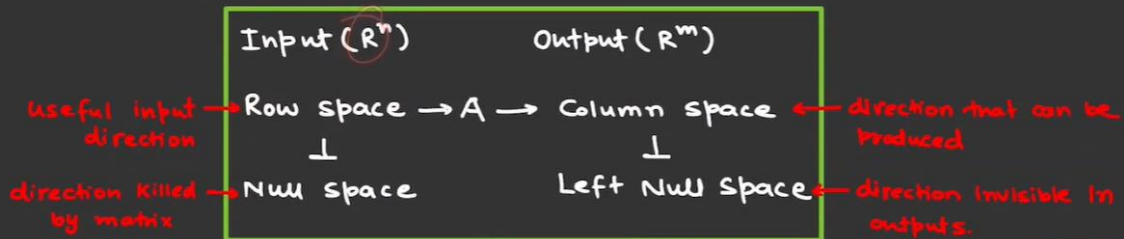
- Null space for matrix A is denoted by $\text{Null}(A)$ or $N(A)$
- Null space of A is the set of all vectors x such that $Ax=0$
 $\downarrow$
 x are in null space
- Null space is a subspace of $\mathbb{R}^n$
- For $A_{m \times n}$ having $\text{Rank}(A)=r$
 $\dim(\text{Null}(A)) = n - r$ ← no of linearly independent vectors for solution of $Ax=0$
 $\uparrow (n-r) = \text{nullity}$

Left Null Space

- Left Null space is denoted by $\text{Null}(A^T)$
 $\uparrow$ Null space of A^T
- It contains all vectors y such that $A^T y = 0$
 $\downarrow$
 y are in Left Null Space
- Null space is a subspace of $\mathbb{R}^m$
- $\dim(\text{Null}(A^T)) = m - r$ ← no of linearly independent vectors for solution of $A^T y = 0$

Linear Transformation Intuition

- $T(x) = Ax$ where $A \in \mathbb{R}^{m \times n}$
 $\uparrow$ Matrix-vector multiplication
- A is a transformation which transform any vector $x \in \mathbb{R}^n$ to $y \in \mathbb{R}^m$



→ Two subspace U and V of $\mathbb{R}^n$ are orthogonal if

↳ every vector in one subspace is perpendicular to every vector in the other.

Example: Discuss all 4 subspace where $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

→ Here A is a linear transformation

$$A: \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$Ax = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Row space ← effective input direction

$$\text{Row}(A) = \text{Span}\left(\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}\right)$$

↳ drop the 3rd coordinate

↳ the x, y plane in input space

Column space ← possible outputs

$$\text{Col}(A) = \text{Span}\left(\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}\right) = \mathbb{R}^2$$

Null space ← killed input direction

$$Ax = 0 \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{Null}(A) = \text{Span}\left(\left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}\right)$$

$$\begin{aligned} x_1 &= 0 \\ x_2 &= 0 \\ x_3 &= t \end{aligned}$$

$$x = t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \leftarrow \text{Z-axis}$$

↳ Z-axis destroyed by transformations.

Left Null space ← forbidden output directions

$$A^T y = 0 \Rightarrow y = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{Null}(A^T) = \{0\} \leftarrow \text{No forbidden direction.}$$